

CSA 0921 - PROGRAMMING IN JAVA

CO2 - AT1 - SCENARIO BASED

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1.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name 'Vidhushini T.S'. The main section is titled 'CO2-AT1- SCENARIOBASED ASSESSMENT'. On the left, a 'QUESTIONS' sidebar lists five questions, with the first one, 'Q1 Student Registration System', selected. The main area displays the scenario: 'Student Registration System' (10 marks). The problem statement asks for three Collection interfaces, a diagram, and a Java program using HashSet and HashMap. The 'Your Code' section shows a Java program that reads student names and courses from input, stores them in a HashMap, and prints the output. The 'Input' section shows the test data: Alice, Java Programming, Data Structures, Bob, Database Systems, Operating Systems. The 'Output' section is currently empty.

```
1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Iterator;
4 import java.util.Map;
5 import java.util.Scanner;
6 import java.util.Set;
7 public class Main {
8     public static void main(String[] args) {
9         Scanner sc = new Scanner(System.in);
10         Map<String, Set<String>> studentMap = new HashMap<>();
11         if (sc.hasNext()) {
12             int studentCount = sc.nextInt();
13             sc.nextLine();
14             for (int i = 0; i < studentCount; i++) {
15                 String studentName = sc.nextLine().trim();
16                 int courseCount = sc.nextInt();
17                 sc.nextLine();
18                 Set<String> courses = new HashSet<>();
19                 for (int j = 0; j < courseCount; j++) {
20                     courses.add(sc.nextLine().trim());
21                 }
22                 studentMap.put(studentName, courses);
23             }
24         } else {
25             Set<String> s1Courses = new HashSet<>();
26             s1Courses.add("Java Programming");
27             s1Courses.add("Data Structures");
28             Set<String> s2Courses = new HashSet<>();
29             s2Courses.add("Database Systems");
30             s2Courses.add("Operating Systems");
31             studentMap.put("Alice", s1Courses);
32             studentMap.put("Bob", s2Courses);
33         }
34         Iterator<Map.Entry<String, Set<String>>> mapIterator = studentMap.entrySet().iterator();
35         while (mapIterator.hasNext()) {
36             Map.Entry<String, Set<String>> entry = mapIterator.next();
37             System.out.println("Student: " + entry.getKey());
38             System.out.print("Courses: ");
39             Iterator<String> courseIterator = entry.getValue().iterator();
40             while (courseIterator.hasNext()) {
41                 System.out.print(courseIterator.next());
42                 if (courseIterator.hasNext()) {
43                     System.out.print(", ");
44                 }
45             }
46             System.out.println();
47         }
48         sc.close();
49     }
50 }
```

This screenshot shows the same SIMATS Online Programming Lab interface, but with the completed code and the resulting output. The 'Your Code' section now includes the logic to initialize the courses for Alice and Bob, and to iterate through the HashMap to print the student names and their respective courses. The 'Output' section displays the results: 'Student: Bob' with courses 'Database Systems, Operating Systems', and 'Student: Alice' with courses 'Data Structures, Java Programming'.

```
11         if (sc.hasNext()) {
12             int studentCount = sc.nextInt();
13             sc.nextLine();
14             for (int i = 0; i < studentCount; i++) {
15                 String studentName = sc.nextLine().trim();
16                 int courseCount = sc.nextInt();
17                 sc.nextLine();
18                 Set<String> courses = new HashSet<>();
19                 for (int j = 0; j < courseCount; j++) {
20                     courses.add(sc.nextLine().trim());
21                 }
22                 studentMap.put(studentName, courses);
23             }
24         } else {
25             Set<String> s1Courses = new HashSet<>();
26             s1Courses.add("Java Programming");
27             s1Courses.add("Data Structures");
28             Set<String> s2Courses = new HashSet<>();
29             s2Courses.add("Database Systems");
30             s2Courses.add("Operating Systems");
31             studentMap.put("Alice", s1Courses);
32             studentMap.put("Bob", s2Courses);
33         }
34         Iterator<Map.Entry<String, Set<String>>> mapIterator = studentMap.entrySet().iterator();
35         while (mapIterator.hasNext()) {
36             Map.Entry<String, Set<String>> entry = mapIterator.next();
37             System.out.println("Student: " + entry.getKey());
38             System.out.print("Courses: ");
39             Iterator<String> courseIterator = entry.getValue().iterator();
40             while (courseIterator.hasNext()) {
41                 System.out.print(courseIterator.next());
42                 if (courseIterator.hasNext()) {
43                     System.out.print(", ");
44                 }
45             }
46             System.out.println();
47         }
48         sc.close();
49     }
50 }
```

Output

```
Student: Bob
Courses: Database Systems,
Operating Systems
Student: Alice
Courses: Data Structures, Java
Programming
```

2.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user's name "Vidhushini T S". The main content area is titled "CO2-AT1- SCENARIOBASED ASSESSMENT". On the left, a "QUESTIONS" sidebar lists five questions, with the first four marked as answered. The selected question, "Q2 E-Commerce Cart System", is displayed in the center. It contains a scenario about an online shopping system and asks for three collection classes, a List-based structure, and a Java program using ArrayList and Iterator. Below the question, the "Your Code" section shows a Java program that implements a shopping cart using ArrayList and Iterator. The "Input" section shows a list of items: Laptop, Headphones, and Mouse. The "Output" section displays a "Security Error: Disallowed code pattern detected." message.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing (10 Marks) 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 import java.util.Scanner;
5 public class Main {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         List<String> cart = new ArrayList<>();
9         if (sc.hasNextInt()) {
10             int addCount = sc.nextInt();
11             sc.nextLine();
12             for (int i = 0; i < addCount; i++) {
13                 cart.add(sc.nextLine().trim());
14             }
15             if (sc.hasNextLine()) {
16                 String removeTarget = sc.nextLine().trim();
17                 Iterator<String> removeIterator = cart.iterator();
18                 while (removeIterator.hasNext()) {
19                     if (removeIterator.next().equalsIgnoreCase(removeTarget)) {
20                         removeIterator.remove();
21                         break;
22                     }
23                 }
24             }
25             System.out.println("Cart Items:");
26             Iterator<String> displayIterator = cart.iterator();
27             while (displayIterator.hasNext()) {
28                 System.out.println(displayIterator.next());
29             }
30             sc.close();
31         }
32     }
33 }
```

Input

3
Laptop
Headphones
Mouse

Output

Security Error: Disallowed code pattern detected.

This screenshot shows the same SIMATS Online Programming Lab interface, but with the completed Java code in the "Your Code" section. The code implements a shopping cart using ArrayList and Iterator, including methods for adding items, removing items, and displaying the cart contents. The "Output" section still shows the "Security Error: Disallowed code pattern detected." message.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing (10 Marks) 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.
(b) Represent how cart items are stored using a List-based structure.
(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 public class Main {
2     public static void main(String[] args) {
3         Scanner sc = new Scanner(System.in);
4         List<String> cart = new ArrayList<>();
5         if (sc.hasNextInt()) {
6             int addCount = sc.nextInt();
7             sc.nextLine();
8             for (int i = 0; i < addCount; i++) {
9                 cart.add(sc.nextLine().trim());
10            }
11            if (sc.hasNextLine()) {
12                String removeTarget = sc.nextLine().trim();
13                Iterator<String> removeIterator = cart.iterator();
14                while (removeIterator.hasNext()) {
15                    if (removeIterator.next().equalsIgnoreCase(removeTarget)) {
16                        removeIterator.remove();
17                        break;
18                    }
19                }
20            }
21            System.out.println("Cart Items:");
22            Iterator<String> displayIterator = cart.iterator();
23            while (displayIterator.hasNext()) {
24                System.out.println(displayIterator.next());
25            }
26            sc.close();
27        }
28    }
29 }
```

Input

3
Laptop
Headphones
Mouse

Output

Security Error: Disallowed code pattern detected.

3.

The screenshot displays the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the page title "SIMATS Online Programming Lab", and the user profile "Vidhushini T S". The main content area is titled "CO2-AT1- SCENARIOBASED ASSESSMENT". On the left, a sidebar lists five questions (Q1 to Q5) with their respective marks. Question Q3, "Library Management System", is selected and shows a 10-mark assessment. The question text describes a library system and asks for three Map implementations, a schema-like representation, and a Java program using HashMap. The "Your Code" section shows a Java program that implements a library management system using HashMap. The "Input" section shows the user's input, and the "Output" section shows the program's output.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 Marks
- Q5 Online Voting System 10 marks

5/5 answered

Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.HashMap;
2 import java.util.Iterator;
3 import java.util.Map;
4 import java.util.Scanner;
5 public class Main{
6     public static void main(String[] args){
7         Scanner sc=new Scanner(System.in);
8         Map<String,String> library=new HashMap<>();
9         if(sc.hasNextLine()){
10             int n=sc.nextInt();
11             sc.nextLine();
12             for(int i=0;i<n;i++){
13                 String isbn=sc.nextLine().trim();
14                 String title=sc.nextLine().trim();
15                 library.put(isbn,title);
16             }
17             if(sc.hasNextLine()){
18                 String searchIsbn=sc.nextLine().trim();
19                 System.out.println("Retrieved Book: "+library.getOrDefault(searchIsbn,""));
20             }
21             else{
22                 library.put("978-0134685991","Effective Java");
23             }
24         }
25     }
26 }
```

Input

```
2
978-0134685991
Effective Java
978-0596009205
```

Output

```
Retrieved Book: Effective Java
All Books in Library:
ISBN: 978-0596009205, Title:
Head First Java
ISBN: 978-0134685991, Title:
```

This screenshot shows the same SIMATS Online Programming Lab interface, but with the completed Java code for the Library Management System. The code is more comprehensive, including the insertion of specific book details and the retrieval of a book by ISBN. The "Output" section shows the program's output, including the retrieved book and a list of all books in the library.

Q3 Online Voting System 10 marks

5/5 answered

Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
5 public class Main{
6     public static void main(String[] args){
7         Scanner sc=new Scanner(System.in);
8         Map<String,String> library=new HashMap<>();
9         if(sc.hasNextLine()){
10             int n=sc.nextInt();
11             sc.nextLine();
12             for(int i=0;i<n;i++){
13                 String isbn=sc.nextLine().trim();
14                 String title=sc.nextLine().trim();
15                 library.put(isbn,title);
16             }
17             if(sc.hasNextLine()){
18                 String searchIsbn=sc.nextLine().trim();
19                 System.out.println("Retrieved Book: "+library.getOrDefault(searchIsbn,""));
20             }
21             else{
22                 library.put("978-0134685991","Effective Java");
23                 library.put("978-0596009205","Head First Java");
24                 String searchIsbn="978-0134685991";
25                 System.out.println("Retrieved book: "+library.getOrDefault(searchIsbn,""));
26             }
27             System.out.println("\nAll Books in Library:");
28             Iterator<Map.Entry<String,String>> iterator=library.entrySet().iterator();
29             while(iterator.hasNext()){
30                 Map.Entry<String,String> entry=iterator.next();
31                 System.out.println("ISBN: "+entry.getKey()+" , Title: "+entry.getValue());
32             }
33             sc.close();
34         }
35     }
36 }
```

Output

```
Retrieved Book: Effective Java
All Books in Library:
ISBN: 978-0596009205, Title:
Head First Java
ISBN: 978-0134685991, Title:
Effective Java
```

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4.

The screenshot displays the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" with a sub-header "JAVA PROGRAMMING". The user is logged in as "Vidhushini T S". The assessment is titled "CO2-AT1- SCENARIOBASED ASSESSMENT".

QUESTIONS

- Q1 Student Registration System (10 marks)
- Q2 E-Commerce Cart System (10 marks)
- Q3 Library Management System (10 marks)
- Q4 Employee Data Processing (10 marks)
- Q5 Online Voting System (10 marks)

5/5 answered

Q4 Employee Data Processing (10 Marks)

Suppose that a company processes employee records and filters employees based on salary.

- List three advantages of Generics in Collections.
- Represent how employee data is stored using List with Generics.
- Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```

import java.util.*;

public class Employee {
    String name;
    int salary;

    Employee(String name, int salary) {
        this.name = name;
        this.salary = salary;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        List<Employee> list = new ArrayList<>();
        for (int i = 0; i < n; i++) {
            list.add(new Employee(sc.next(), sc.nextInt()));
        }
        Iterator<Employee> it = list.iterator();
        while (it.hasNext()) {
            Employee e = it.next();
            if (e.salary > 50000) {
                System.out.println(e.name + " " + e.salary);
            }
        }
    }
}

```

Input

```

2
Arun 45000
Sala 60000

```

Output

```

Sala 60000

```

Run Save

36°C Mostly cloudy 14:51 01-08-2026

5.

The screenshot displays the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" with a sub-header "JAVA PROGRAMMING". The user is logged in as "Vidhushini T S". The assessment is titled "CO2-AT1- SCENARIOBASED ASSESSMENT".

QUESTIONS

- Q1 Student Registration System (10 marks)
- Q2 E-Commerce Cart System (10 marks)
- Q3 Library Management System (10 marks)
- Q4 Employee Data Processing (10 marks)
- Q5 Online Voting System (10 marks)

5/5 answered

Q5 Online Voting System

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- Identify three suitable Collection types for this system.
- Design a structure using Set and Map for voters and vote counting.
- Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting
 - Display results

Your Code

```

import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Set<String> voters = new HashSet<>();
        Map<String, Integer> voteCounts = new HashMap<>();
        if (sc.hasNextLine()) {
            int totalVotes = sc.nextInt();
            sc.nextLine();
            for (int i = 0; i < totalVotes; i++) {
                String line = sc.nextLine().trim();
                String[] parts = line.split("\\s+");
                if (parts.length >= 2) {
                    String voterId = parts[0];
                    String candidate = parts[1];
                    if (voters.contains(voterId)) {
                        System.out.println("Duplicate vote attempt by: " + voterId);
                    } else {
                        voters.add(voterId);
                        voteCounts.put(candidate, voteCounts.getOrDefault(candidate, 0) + 1);
                        System.out.println("Vote recorded for " + candidate);
                    }
                }
            }
        }
    }
}

```

Input

```

4
V101 Alice
V102 Bob
V101 Alice
V103 Alice

```

Output

```

Vote recorded for Alice by V101
Vote recorded for Bob by V102
Duplicate vote attempt by V101
Vote recorded for Alice by V103

```

Run Save

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SEARCH

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SIMATS
ENGINEERING

SIMATS Online Programming Lab

Vidhushini T S

REGISTRATION

JAVA PROGRAMMING

```
13 int totalVotes=0,nextLine();
14 sc.nextLine();
15 for(int i=0;i<totalVotes;i++){
16     String line=sc.nextLine().trim();
17     String[] parts=line.split("\\s+");
18     if(parts.length>2){
19         String voterId=parts[0];
20         String candidate=parts[1];
21         if(votedUsers.containsKey(voterId)){
22             System.out.println("Duplicate vote attempt by: "+voterId);
23         }else{
24             votedUsers.add(voterId);
25             voteCounts.put(candidate,voteCounts.getOrDefault(candidate,0)+1);
26             System.out.println("Vote recorded for "+candidate);
27         }
28     }
29 }
30 }else{
31     String[][] sampleVotes={
32         {"V001","Alice"},
33         {"V002","Bob"},
34         {"V003","Alice"},
35         {"V004","Alice"}
36     };
37     for (String[] vote:sampleVotes){
38         String voterId=vote[0];
39         String candidate=vote[1];
40         if(votedUsers.containsKey(voterId)){
41             System.out.println("Duplicate vote attempt by: "+voterId);
42         }else{
43             votedUsers.add(voterId);
44             voteCounts.put(candidate,voteCounts.getOrDefault(candidate,0)+1);
45             System.out.println("Vote recorded for "+candidate);
46         }
47     }
48 }
49 System.out.println("\n--- Final Voting Results ---");
50 IteratorMap.Entry<String,Integer> iterator=voteCounts.entrySet().iterator();
51 while(iterator.hasNext()){
52     Map.Entry<String,Integer> entry=iterator.next();
53     System.out.println("Candidate: "+entry.getKey()+" | Votes: "+entry.getValue());
54 }
55 }
56 }
57 }
```

36°C

Mostly cloudy



ENG

IN

14:52

01-08-2026